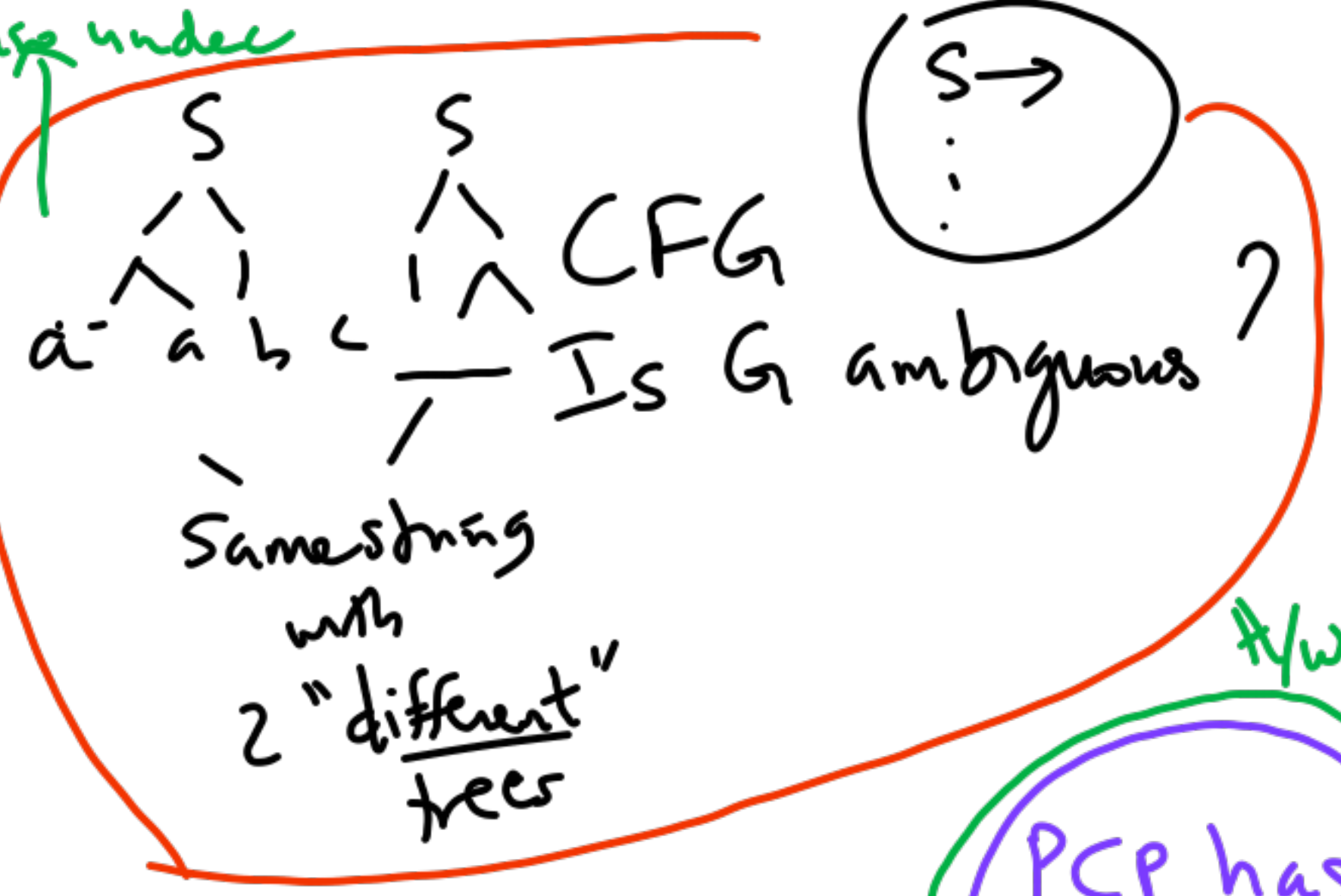
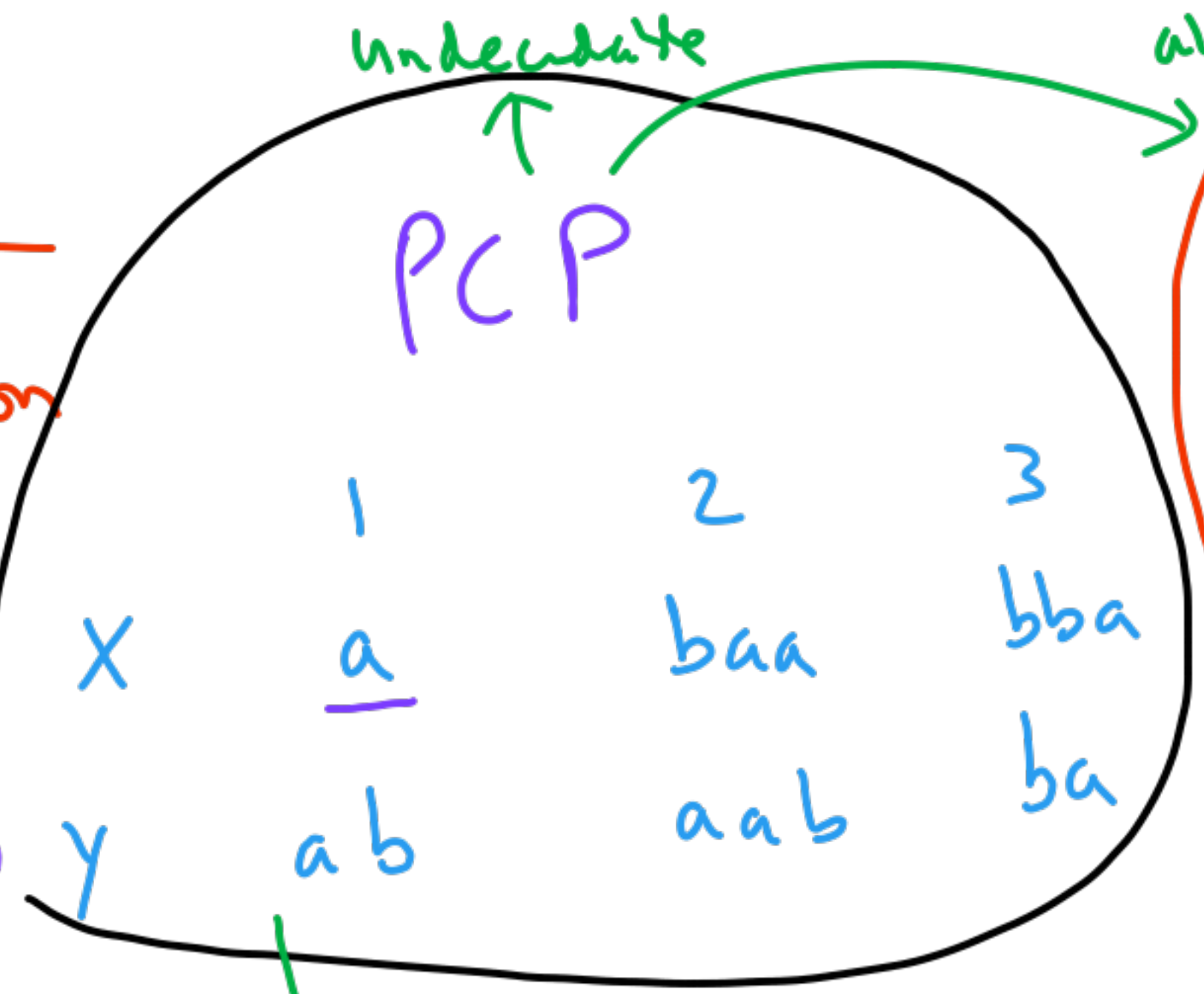


3.10.2025

① Problem Reduction

Example

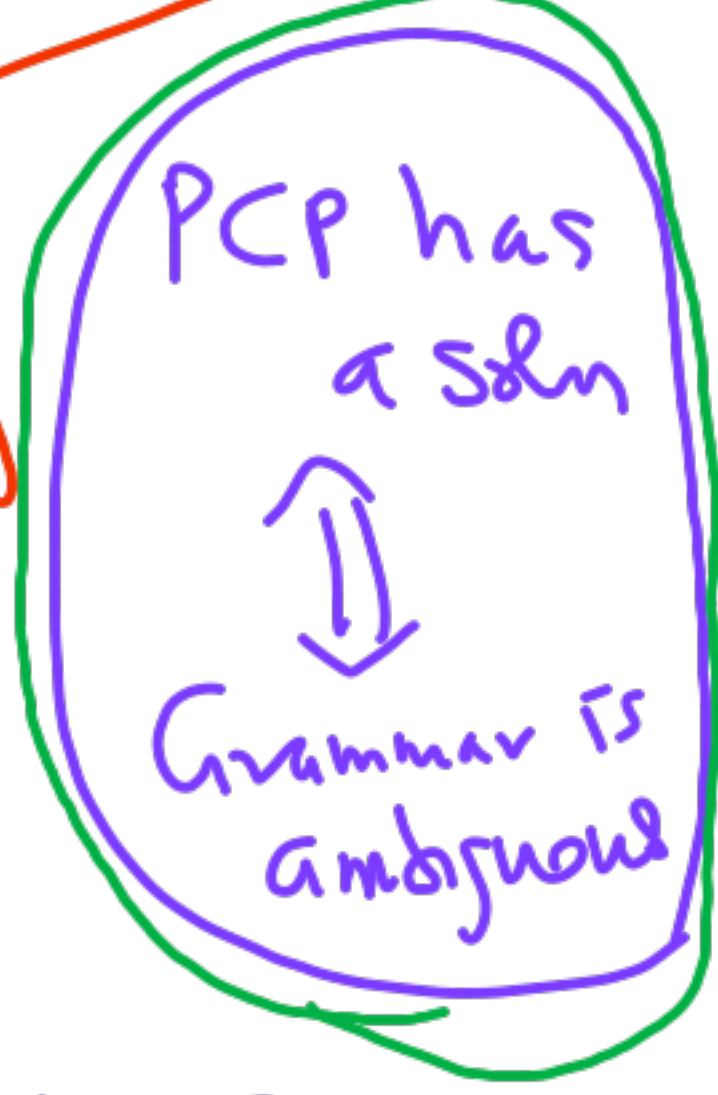
A →
B →



#/w

PCP → "Convert" → Gr. Ambiguity

"cost"



Soln

1 2* 3

1 3

abba
abba
1 3

$S \rightarrow A \mid B$

(Equivalent for decidability)

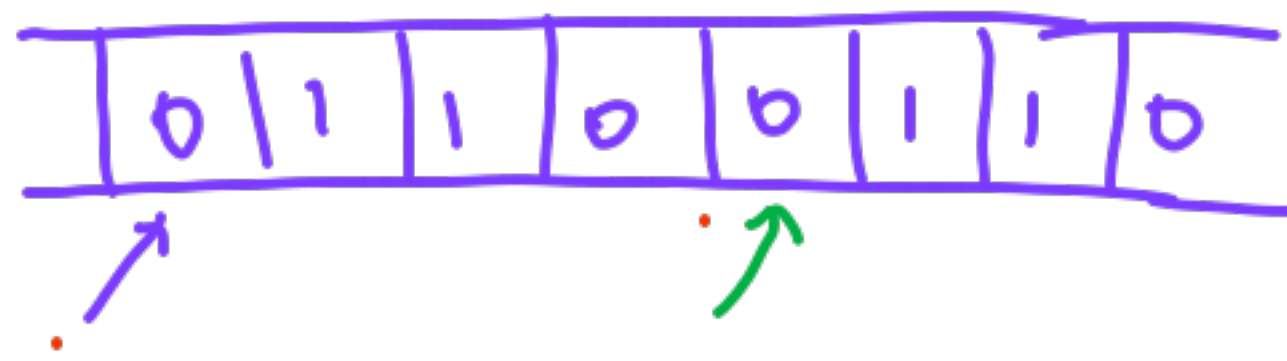
$A \rightarrow aA1 \mid a1 \mid baaA2 \mid baa2 \mid bbaA3 \mid bba3$

$B \rightarrow abB1 \mid ab1 \mid aabB2 \mid aab2 \mid baB3 \mid ba3$

2-7 wpe

Non-det TM

for $L = \{ww \mid w \in (01)^*\}$

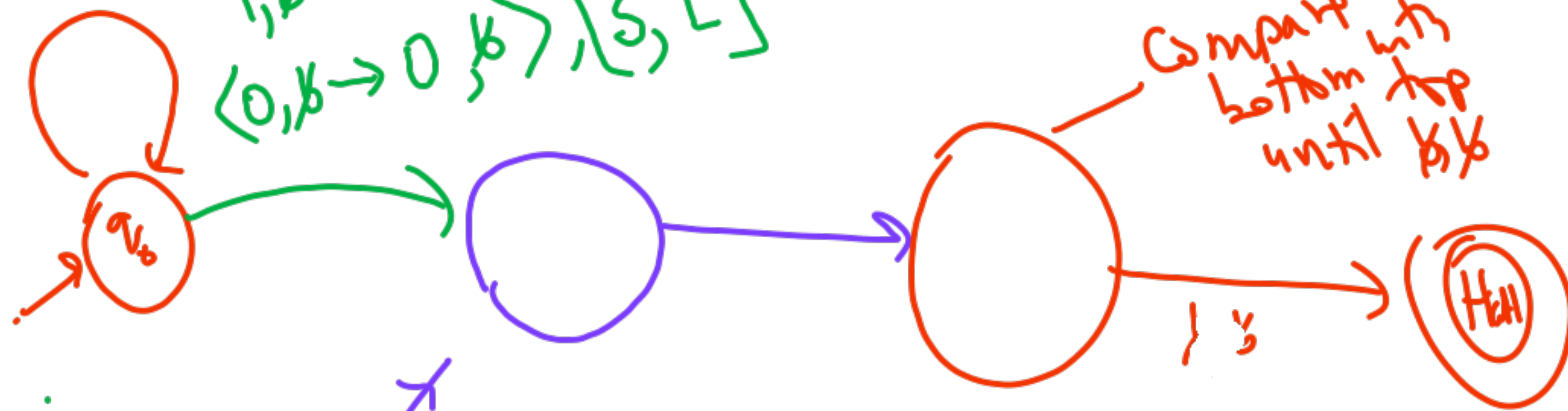


0 1 1 0

$\langle 1, b \rightarrow 1, D \rangle$
 $\langle 0, b \rightarrow 0, 0 \rangle, [R, R]$

$1, b \rightarrow 1, b$
 $\langle 0, b \rightarrow 0, b \rangle, [S, L]$

Compare
bottom with
top
until b/b



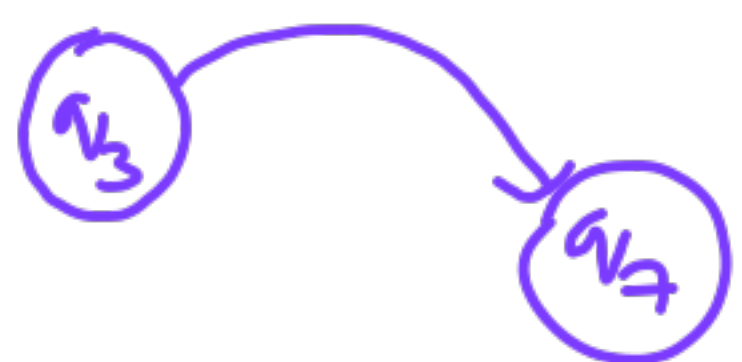
Keep going left on bottom tap until b

Extension $\rightarrow$ Stay

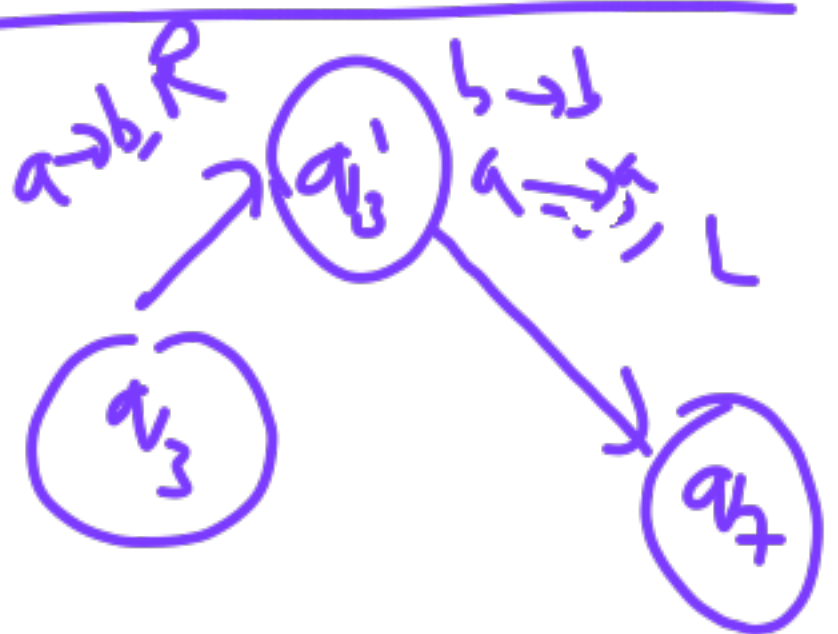
$L, R \rightarrow$ stay $\rightarrow$ non-det

L, R, S
stay

$a \rightarrow b, S$



$x \rightarrow x$



Decidability

All languages $L \subseteq \Sigma^*$

Rec. Enumerable

"Semi-Decidable"

Rec. Decidable

L_d

L_d

Antenna

Diagrammed

L is decidable if

$w \rightarrow M \rightarrow \begin{cases} \text{Yes if } w \in L \\ \text{No if } w \notin L \end{cases}$

and M always halts.

L is r.e (semi-decidable) if there is a TM, M

$w \rightarrow M \rightarrow \begin{cases} \text{Yes if } w \in L \text{ \& halt} \\ \text{No \& halt} \\ \text{Run forever} \end{cases} w \notin L$

L_d fails here also

Is G ambiguous?

Does PC have a solution

Hilbert's 10th problem

$$7x^3y^2 - 11xy^2z - \dots = 0?$$

not recursive

① TMs are enumerating Countable

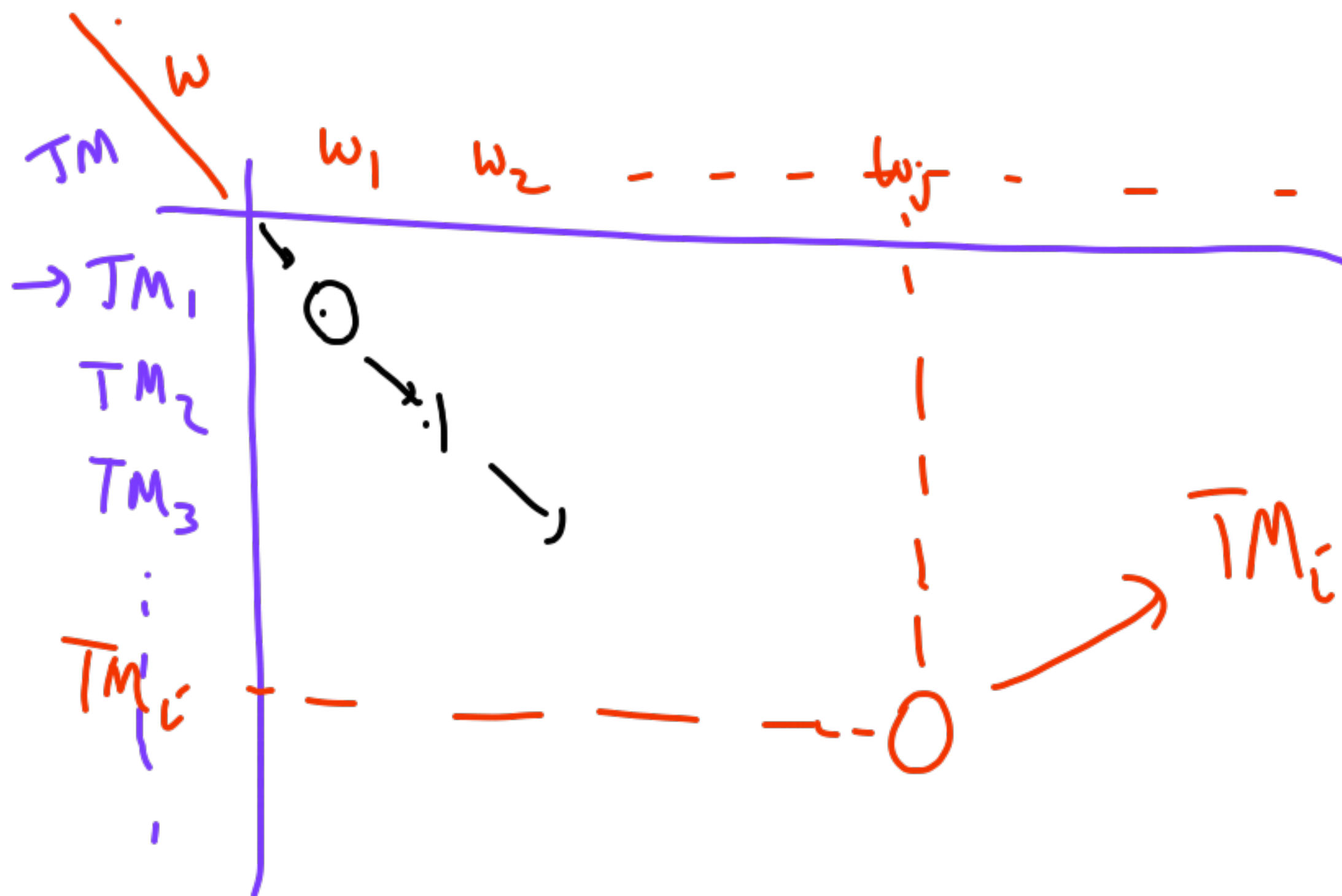
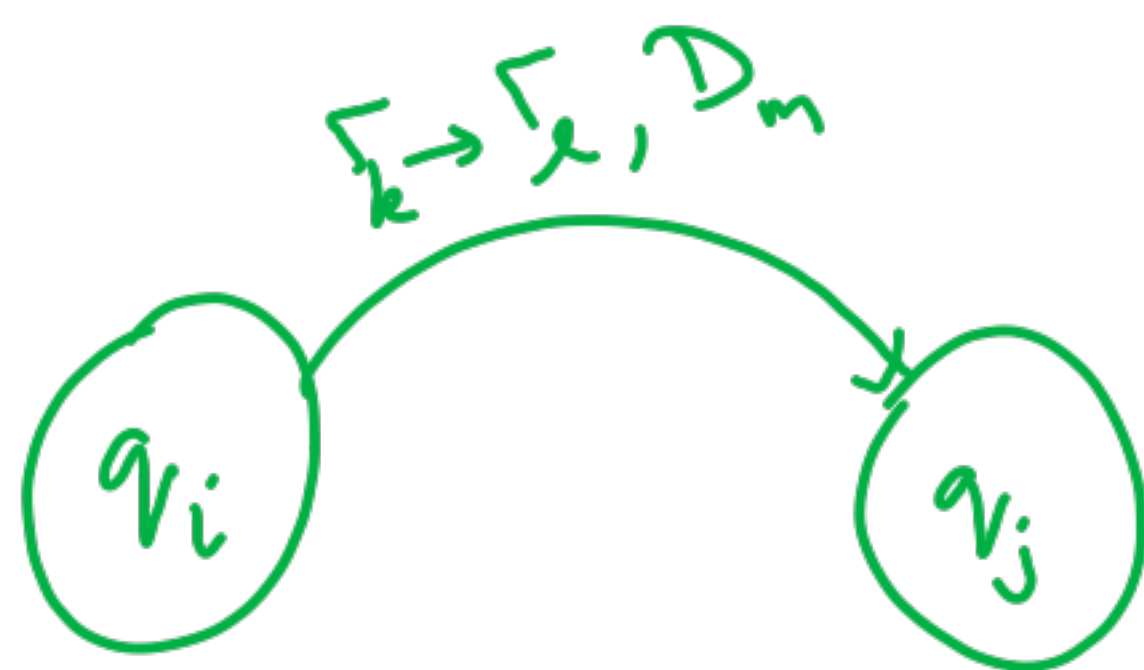
② there is a language L_d no TM can accept.

③ there is a language L_w that a TM can enumerate but cannot "decide"

1-move

$0^i | 0^j | 0^k | 0^l | 0^m$

$\underbrace{\text{move}_1}_{\text{2 ones}} \parallel \underbrace{\text{move}_2}_{\text{2 ones}} \parallel \dots \parallel \text{move}_n$

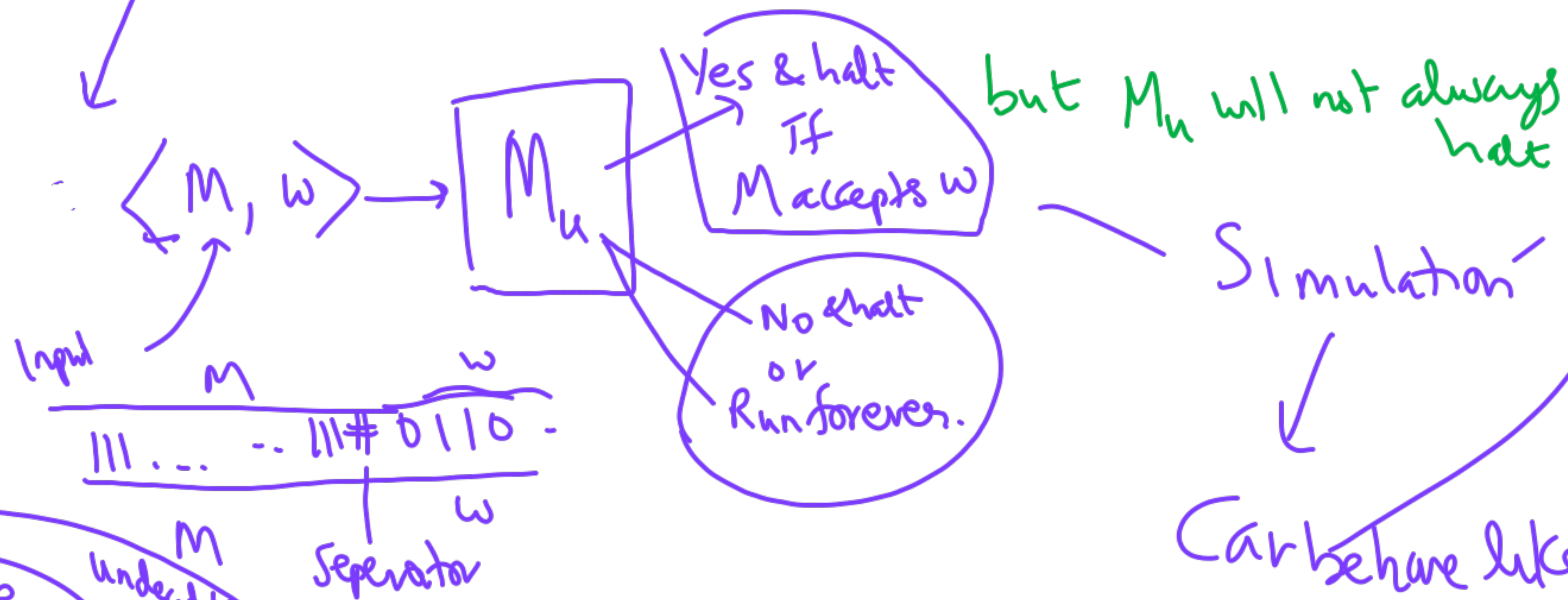


$$L_d = \left\{ w \mid \begin{array}{l} w = w_i \\ \text{and } TM_i \text{ does not} \\ \text{accept } w_i \end{array} \right\}$$

cannot be accepted by any TM

Proof by contradiction - if $\bar{T}_k$ accepts L_d then $\bar{T}_k$ gives wrong answer on w_k

L_u is a language for which we can build a TM M_u universal TM



Can behave like any other TM

